PG – 298

III Semester M.C.A. Examination, April/May 2022
(CBCS Y2K20 Scheme)
COMPUTER SCIENCE
3MCA3 : Research Methodology

Time : 3 Hours Max. Marks : 70

Instructions *: 1) Answer* ***any five*** *questions from Section – **A**, **each** question carries **6** marks.*
2) Answer ***any four full*** *questions from Section – **B**, **each** question carries **10** marks.*

SECTION – A **(5×6=30)**

1. Define Research. How to frame title of Research ? Explain with examples.
2. All student in a class have age 20. What is the standard deviation ? How to interpret the data ?
3. What is the use of computer in research ?
4. Explain data analysis with GRETL tool.
5. Compare correlation and regression models.
6. Explain optimization using fuzzy system.
7. Summarize the outline of the popular report.
8. Discuss any three phases of genetic algorithm.

SECTION – B **(4×10=40)**

9. Explain various steps of research process.
10. How to test a hypothesis ? Explain the chi-square test steps with example.
11. Explain how statistical measures of central tendency used in research.
12. Discuss any one time series model and its use.
13. Explain the role of simulation in research with example.
14. Explain the mechanism of writing a research report.

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JP – 770

III Semester M.C.A. (Two Years Course) Examination, May/June 2023
(CBCS 2021 – 22)
3MCA3 : RESEARCH METHODOLOGY

Time : 3 Hours Max. Marks : 70

SECTION – A

Answer **any five** questions. **(5×6=30)**

1. Define and explain the term "Research". Explain its importance and objectives.
2. Briefly explain the significance of data processing. What is the problem associated with data processing ?
3. Discuss various methods of research. Explain various factors to be considered in making the decision of sampling method.
4. Why literature review is necessary for research ? Explain list and mention various steps involved in conducting a literature review.
5. What is time series analysis in Research Methodology ? Discuss the major components of time series analysis.
6. What is correlation and types of correlation ? Give examples of variables in the medical field that are likely to be correlated. Give examples of variables that are positively correlated and variables that are negatively correlated.
7. Explain the steps involved in research design.
8. Justify how probability sampling is better than non probabilistic sampling.

SECTION – B

Answer **any four** questions. **(4×10=40)**

9. Define the main issues which should receive the attention of the researcher in formulating the research problem. Give suitable example.
10. What are the ethical issues concerning the research participants of a research activity ? Explain with a case study.

P.T.O.

11. Explain the different steps involved in writing a research report.

12. Define soft computing technology and list some techniques. Compare soft computing and hard computing.

13. Write a short note on :

    1) Fuzzy logic. 5

    2) Genetic algorithm. 5

14. Write the concept of "Plagiarism". List out some of Plagiarism detection tools available.

PG – 373

III Semester M.C.A. (Two Years Course) Examination, May 2024
(CBCS) (2021-22)
COMPUTER SCIENCE
3MCA3 : Research Methodology

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any 5** from Section – **A**.*
*2) Answer **any 4** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Explain principles of the APA Code of Ethics.
2. Discuss all the steps in research process.
3. Explain correlation and types of correlation.
4. Discuss resampling techniques used during data exploration.
5. Explain the rules of graphical representation of data.
6. Discuss bar graph with a suitable example.
7. What is soft computing ? Discuss various soft computing techniques.
8. Discuss about the process involved in writing a research report.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. Discuss in detail about different types of research.
10. Explain time series analysis in detail.

P.T.O.

11. Discuss in detail about classification as a statistical modelling technique with example.

12. Explain tabular description of data in detail.

13. Explain in detail about Statistical Package for the Social Sciences (SPSS) and GRETL Tools.

14. Write short notes of the following :

    a) Structure and components of the research report.

    b) Referencing in academic writing.

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PG – 669

III Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS)(2021-22)
COMPUTER SCIENCE
3MCA3 : Research Methodology

Time : 3 Hours Max. Marks : 70

Answer **any five** questions. **Each** question carries **6** marks. **(6×5=30)**

1. What is Research ? Describe its types and significance in modern times.
2. "A research scholar has to work as a judge and derive the truth and not as a pleader who is only eager to prove his case in favour of his plaintiff". Discuss the statement pointing out the objectives of research.
3. Explain the process of identification of research problem.
4. What is literature review ? How do you carry out literature review ? What are the advantages of the same ?
5. Discuss the probability sampling with merits and demerits.
6. What is hypothesis ? What characteristics it must possess in order to be a good research hypothesis ?
7. "A sample may be large yet worthless because it is not random; or it may be random but unreliable because it is small." Comment upon the above statement and explain the importance of sampling in daily life.
8. What is soft computing ? Differentiate between soft computing and hard computing ?

Answer **any four** questions. **Each** question carries **10** marks. **(10×4=40)**

9. "Research is much concerned with proper fact finding, analysis and evaluation". Do you agree with this statement ? Give reason in support of your answers. Also describe the different steps involved in a research process.
10. What is statistical modeling and analysis ? Discuss about time series analysis.

P.T.O.

11. Explain the difference between the mean and median. Provide examples to illustrate situations where each measure is more appropriate in representing central tendency. Highlight the advantages and limitations of using mean and median in data analysis.

12. Discuss the characteristics and applications of a histogram and a frequency polygon. Explain how each type of graphical representation displays frequency distribution patterns. Provide examples of datasets that would be effectively represented by each type and explain the benefits of using each in data visualization.

13. Find the correlation coefficient between plant height and number of pads.

X	15	20	17	22	25	29	12
Y	18	17	21	23	20	19	22

14. Write a short note on :
    a) Ethics in Research. **5**
    b) ANN. **5**

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PG – 672

III Semester M.C.A. (Two Years Course) Examination, May/June 2026
(CBCS) (2021 – 22)
COMPUTER SCIENCE
3MCA3 : Research Methodology

Time : 3 Hours Max. Marks : 70

Instructions : 1) Answer ***any five*** questions from Section – ***A***.
2) Answer ***any four*** questions from Section – ***B***.

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define research. Explain the types of research with examples.
2. What is a research problem ? How do you formulate a research problem ? Explain.
3. Differentiate between correlation and regression.
4. Explain different types of graphs used in data representation.
5. Define data preparation. Discuss steps in data preparation.
6. Explain the role of computers in research.
7. Describe referencing styles in academic writing.
8. Describe frequency distribution and its types.

SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. Explain the research process in detail with a diagram.
10. What is soft computing ? And discuss the different soft computing algorithms. Explain Fuzzy logic.

P.T.O.

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11. Calculate Karl Pearson's coefficient of correlation for the data :

X	10	20	30	40	50
Y	15	25	35	45	55

i) Find mean of X and Y

ii) Compute deviations

iii) Calculate correlation coefficient.

12. What is Time series analysis in research ? Discuss different types of Time series components.

13. Discuss genetic algorithms and their working principles.

14. Explain in detail the structure and components of a research report.

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